

Maulana Azad National Institute of Technology, Bhopal
End-Term Examination, November 2024

Department: Electrical Engineering

Subject Name: Measurement and Instrumentation

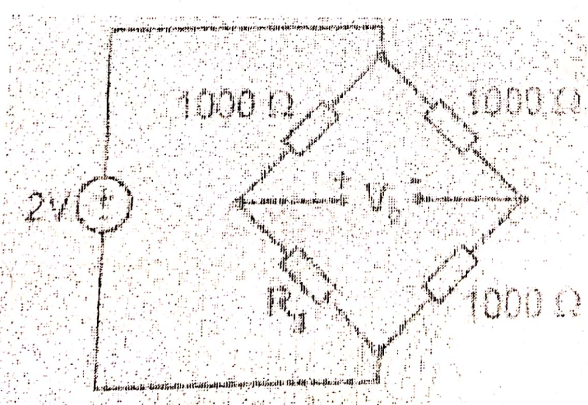
Max. Marks: 50

Subject Coordinators Name: Dr. Sushma Gupta, Dr. Punjan Dohare and Rakeshwri Agrawal

Semester: B. Tech. III

Subject Code: EE-214

Time: 3.00 Hrs

Qu. No.	Question	Marks	COs
1 (a)	What is drift? Explain the different types of drifts with sketches of input-output relationship in each case. How errors in measurement can be classified? A voltmeter reads 111.5 V. The error taken from an error curve is 5.3%. Find the true value of the voltage.	(5)	CO1
(b)	What is the importance of controlling torque in analog instruments? How it is provided in instrument? Explain the working operation for moving iron instruments & derive it's torque equation.	(5)	CO2
2 (a)	Why scale of moving iron instruments is non-uniform while that of PMMC instruments is uniform. The inductance of a moving iron ammeter with a full scale deflection of 90° at 1.5A, is given by the expression $L = (200 + 40\theta - 4\theta^2 - \theta^3) \mu\text{H}$, where θ is the deflection in radian from the zero position. Estimate the angular deflection of the pointer for a current of 1.0 A.	(5)	CO2
(b)	Why PMMC instruments work only on DC not AC? With neat diagram explain the construction, working, torque equation and advantages, disadvantages of PMMC instrument.	(5)	CO2
3 (a)	How to measure liquid level using inductive and capacitive transducers methods?	(5)	CO3
(b)	(i) The LVDT core is slotted longitudinally to reduce----- (ii) The LVDT is placed-----to protect from environmental variation. (iii) The gauge factor of a strain gauge would be----- if the value of resistance is 152 Ω for 5000 microstrain. (iv) A 1000 Ω strain gauge has a gauge factor of 2.5. It is connected in the bridge as shown in fig. The strain gauge is subjected with a positive strain of 400 $\mu\text{m/m}$. The output V_o (in mV) of the bridge is -----.  (v) The reading of the pressure gauge fitted on a vessel is 25 bar. The atmospheric pressure is 1.03 bar and the value of g is 9.81 m/s². The absolute pressure in the vessel is -----.	(5)	CO4

4 (a)	Which type of temperature transducer is being preferred if one wants to measure temperature between -60°C to 15°C . Describe in detail which being used.	(3)	CO3																				
(b)	Draw the voltage-current and current-time characteristics of Thermistor and explain it.	(3)	CO3																				
(c)	Use the following values of resistance versus temperature for an RTD to find the linear and Quadratic approximation of resistance between 100°C and 130°C about a mean temperature of 115°C . <table><tr><td>Temp $^{\circ}\text{C}$</td><td>90</td><td>95</td><td>100</td><td>105</td><td>110</td><td>115</td><td>120</td><td>125</td><td>130</td></tr><tr><td>Resistance Ω</td><td>562.66</td><td>568.03</td><td>573.40</td><td>578.77</td><td>584.13</td><td>589.48</td><td>594.84</td><td>600.18</td><td>605.52</td></tr></table>	Temp $^{\circ}\text{C}$	90	95	100	105	110	115	120	125	130	Resistance Ω	562.66	568.03	573.40	578.77	584.13	589.48	594.84	600.18	605.52	(4)	CO4
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Resistance Ω	562.66	568.03	573.40	578.77	584.13	589.48	594.84	600.18	605.52														
5 (a)	Discuss the major sources of errors and their effects on the performance of current transformer. Describe the methods to reduce the errors in current transformer.	(5)	CO5																				
(b)	A potential transformer rated 13800/230 V has 24000 turns in the primary winding and 400 turns in the secondary winding. With 13800 V applied to the primary and the secondary circuit open circuited, the primary winding current is 0.01A, lagging the voltage by 73.7° . With a particular burden connected to the secondary, the primary winding current is 0.025A, lagging the voltage by 53.1° . Primary winding resistance = $1200\ \Omega$ Primary winding reactance = $2000\ \Omega$ Secondary winding resistance = $0.5\ \Omega$ Secondary winding reactance = $0.8\ \Omega$ i. Find the secondary current, terminal voltage and load burden, when voltage $V_p = 13800 + j0$ is applied to primary. ii. Find the actual transformation ratio and the phase angle.	(5)	CO5																				